

Week 4 Learning Behaviors

Theme

Over the last three weeks, you have built systems where behavior was defined by rules.

Examples include:

- Conway's Game of Life
- Neural Cellular Automata
- Boids
- Information Propagation
- Zombie Evacuation

In all of these systems, the rules were written by a human.

This week asks a different question:

What happens when we stop writing rules and start learning them?

The objective is not to learn deep learning theory.

The objective is to understand how learning can be inserted into an emergent system.

Learning Goals

By the end of this week, you should be able to:

- Identify hardcoded rules in a system.
 - Decide which rules should remain fixed.
 - Decide which rules can be learned.
 - Design an agent that learns from observations.
 - Connect neural networks to the simulations you have already built.
 - Reason about the behaviors that might emerge from learned policies.
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Part 1 Analyze Existing Systems

Choose one of the systems you built in previous weeks:

- Boids
- Zombie Evacuation
- Information Spread
- Conway's game of life

For that system answer:

What rules are hardcoded?

Examples:

- Move away from zombies.
 - Align with neighbors.
 - Spread information within radius.
 - Update cells according to fixed logic.
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Which rules could be learned?

Examples:

Instead of:

"Move away from zombie"

learn:

Observation → Action

Instead of:

"Align with neighbors"

learn:

Local State → Velocity

Part 2 Design a Learnable World

Design a new simulation.

Possible examples:

- Zombie outbreak
- Trust network
- Survivor camp
- Resource economy
- Robot swarm
- Predator-prey ecosystem

You may propose your own.

Define the Observation Space

What information can the agent observe?

Examples:

- Distance to zombie
- Distance to exit
- Nearby agents
- Available resources
- Trust score

Important:

Agents should only have local information.

Define the Action Space

What actions can the agent take?

Examples:

- Move
- Hide
- Cooperate

- Defect
 - Share information
 - Collect resources
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Define Success

How will the agent know if it is doing well?

Examples:

- Stay alive
 - Reach safety
 - Maximize resources
 - Maintain trust
 - Protect teammates
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Part 3 Replace a Rule with Learning

Choose at least one rule and replace it with a neural network.

Examples:

Learn Boid Steering

Input:

- Neighbor positions
- Neighbor velocities

Output:

- Movement direction
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Learn Zombie Escape

Input:

- Nearby zombies

- Nearby civilians
- Distance to exit

Output:

- Movement direction
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Learn Trust Decisions

Input:

- Trust score
- Previous interactions
- Nearby agents

Output:

- Cooperate
 - Defect
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Part 4 Predict Emergent Behavior

Before running your experiment answer:

What behaviors do you expect to emerge?

Examples:

- Flocking
 - Trust clusters
 - Panic waves
 - Resource monopolies
 - Survivor groups
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Why?

Explain how local observations and local decisions could create these global behaviors.

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